01. Feasify System Analysis and Architectural Specification

Project Name: Feasify Capstone System

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Auditor: QA Engineering & Architecture Review Team

Evaluation Target: Production Codebase (`Feasify/` & `feasify-backend/`)

1. Executive Summary

Feasify is an academic web-based capstone proposal and feasibility evaluation management system tailored for Philippine universities. The platform establishes collaborative workflows between three core user roles: **Students**, **Faculty Advisers**, and **Academic Chairpersons**. Feasify guides students through business proposal ideation, Philippine government registration name checks (DTI BNRS & SEC eSPARC), location mapping, 5-year financial statement projections (incorporating BMBE Act 3% smart tax vs. CREATE Act 20% CIT), and automated AI feasibility scoring powered by Google Gemini.

2. Actual Technology Stack Identification

Based on exhaustive static inspection of `Feasify/package.json` and `feasify-backend/package.json`, the verified technology stack is:

Layer	Component / Library	Version	Verified Purpose
Frontend Framework	React	`^19.2.0`	Declarative reactive UI components
Frontend Language	TypeScript	`~5.9.3`	Type-safe client-side application logic
Build & Dev Tool	Vite	`^7.3.1`	Ultra-fast ESM frontend bundling and HMR
Styling & Design	Tailwind CSS & PostCSS	`^4.2.0`	Utility-first responsive design tokens
Routing	React Router DOM	`^7.13.0`	Client-side routing with role-based `ProtectedRoute`
Database & Auth (Client)	Firebase JS SDK	`^12.9.0`	Firebase Auth, Cloud Firestore client, Storage
Real-Time Messaging	Socket.IO Client	`^4.8.3`	Low-latency collaborative team messaging
Mapping Engine	Leaflet	`^1.9.4`	OpenStreetMap client with custom Philippine landmarks
Charts & Data Viz	Recharts	`^3.8.1`	Financial break-even, sensitivity, and radar charts
Export Engines	jsPDF, html2canvas, SheetJS (`xlsx`)	Latest	PDF generation, table rasterization, Excel exports
Backend Runtime	Node.js (CommonJS)	`>=18.0.0`	Backend asynchronous event-driven server runtime
Backend Framework	Express	`^5.2.1`	REST API routing and middleware management
Backend Database/Auth	Firebase Admin SDK	`^14.3.0`	Privileged Firestore access & ID token verification
AI LLM Engine	`@google/generative-ai`	`^0.21.0`	Multi-model cascade (`gemini-3.5-flash-lite`)
Traffic Control	`express-rate-limit`	`^8.6.2`	IP & token-based rate limiting (60 req/min)

3. High-Level Architecture Diagrams

3.1 C4 Container Diagram

```
graph TD
    subgraph ClientLayer ["Client Layer (React 19 + TypeScript + Vite)"]
        UI_Student["Student Portal<br/>(Projects, Financials, AI Analysis)"]
        UI_Adviser["Adviser Portal<br/>(Dashboard, Evaluation, Section Rules)"]
        UI_Chairperson["Chairperson Portal<br/>(Users, Projects, Audit Trail)"]
        ClientAuth["Firebase Auth Client"]
        ClientFirestore["Firestore Client SDK"]
        ClientSocket["Socket.IO Client"]
    end

    subgraph BackendLayer ["Backend Engine (Node.js + Express 5)"]
        Server["Express 5 REST API (:10000)"]
        TokenMW["verifyToken Middleware"]
        LimiterMW["Rate Limiters (60 req/min)"]
        NameChecker["NameCheckerService<br/>(DTI & SEC Adapters)"]
        GeminiCascade["Gemini Cascade Service"]
        ExemplarEngine["Dynamic Exemplars Engine"]
        SocketServer["Socket.IO Server"]
        AdminSDK["Firebase Admin SDK"]
    end

    subgraph DataServices ["External & Cloud Services"]
        FirebaseAuthService[("Firebase Authentication")]
        CloudFirestore[("Cloud Firestore Database")]
        GoogleAI[("Google Gemini LLM API")]
        DTI_BNRS[("DTI BNRS Portal / Search")]
        SEC_ESPARC[("SEC eSPARC Portal")]
        OSM_Nominatim[("OpenStreetMap / Nominatim")]
    end

    UI_Student --> ClientAuth
    UI_Adviser --> ClientAuth
    UI_Chairperson --> ClientAuth
    ClientAuth --> FirebaseAuthService

    UI_Student --> ClientFirestore
    UI_Adviser --> ClientFirestore
    UI_Chairperson --> ClientFirestore
    ClientFirestore --> CloudFirestore

    UI_Student --> ClientSocket
    ClientSocket <--> SocketServer

    UI_Student --> Server
    UI_Adviser --> Server
    Server --> TokenMW
    Server --> LimiterMW
    Server --> NameChecker
    Server --> GeminiCascade
    Server --> ExemplarEngine
    Server --> AdminSDK

    AdminSDK --> CloudFirestore
    GeminiCascade --> GoogleAI
    NameChecker --> DTI_BNRS
    NameChecker --> SEC_ESPARC
```

3.2 End-to-End Proposal Data Flow (DFD Level 1)

```
sequenceDiagram
    autonumber
    actor Student as Student Leader
    participant FE as Feasify Client (Projects.tsx)
    participant BE as Express Backend (:10000)
    participant FS as Cloud Firestore
    actor Adviser as Faculty Adviser

    Student->>FE: Fill 11 required proposal fields
    FE->>BE: GET /api/teams/:groupId/proposals/count
    BE-->>FE: { count: N, max: 3, allowed: boolean }

    alt Proposals Count >= 3
        FE-->>Student: Modal: "Proposal Limit Reached (Max 3 Allowed)"
    else Proposals Count < 3
        Student->>FE: Click "Submit to Adviser"
        FE->>FS: addDoc(collection(db, "proposals"), proposalData)
        FS-->>FE: DocRef created (prop_id)
        FE->>FS: updateDoc(groups/:groupId, status: "Pending Review")
        FE->>FS: addDoc(notifications, { to: adviserId, type: "proposal" })
        FE-->>Student: Toast: "Proposal submitted successfully!"

        Note over Adviser,FS: Real-time onSnapshot triggers in AdviserDashboard
        FS-->>Adviser: Live Proposal update received
        Adviser->>BE: PUT /api/proposals/:proposalId/status (Approved/Rejected)
        BE->>FS: updateDoc(proposals/:proposalId, { status, adviserRemarks })
        FS-->>FE: onSnapshot updates student dashboard status badge
    end
```

4. Feature Inventory & Implementation Matrix

Module	Purpose	Key Files	DB Collection	API Endpoint	Roles	Implementation Status
Authentication	Registration, login, password reset, session persistence	`Auth.tsx`, `ResetPassword.tsx`, `firebase.ts`	`users`	Client SDK (`fbAuth`)	All	**Implemented**
Role Routing	RBAC route authorization and redirect	`ProtectedRoute.tsx`, `App.tsx`	`users`	Client Firestore lookup	All	**Implemented**
Proposal Creation	Business idea draft and submission	`Projects.tsx`, `proposalDetails.ts`	`proposals`, `groups`	Direct Firestore + `/api/teams/:id/proposals`	Student	**Implemented**
3-Proposal Limit	Enforce max 3 proposals per capstone team	`Projects.tsx`, `feasify-backend/index.js`	`proposals`	`/api/teams/:groupId/proposals/count`	Student	**Implemented**
Name Checker	DTI BNRS & SEC eSPARC name conflict check	`OfficialNameChecker.tsx`, `services/nameChecker/`	`companyNames.json`, `copyright_db.json`	`POST /api/name-check`	Student, Adviser	**Implemented**
Location Mapping	Philippine location selection and pin drop	`LocationPickerMap.tsx`, `philippinePlaceSearch.ts`	Stored in `proposals.proposedLocation`	Nominatim / Local DB	Student	**Implemented (Leaflet/OSM)**
Financial Analysis	5-year statements, BMBE vs CREATE tax	`Financial_input.tsx`, `balanceSheetProjections.ts`	`proposals`, `financials`	`POST /api/analyze`	Student	**Implemented**
AI Feasibility	Automated Gemini feasibility audit & rubric	`AI_Analysis.tsx`, `AdviserAIRules.tsx`	`proposals`	`POST /api/analyze-financials`, `/api/analyze-proposal`	Student, Adviser	**Implemented**
Adviser Review	Review proposals, assign statuses, give remarks	`AdviserDashboard.tsx`	`proposals`, `groups`	`PUT /api/proposals/:id/status`	Adviser	**Implemented**

Module	Purpose	Key Files	DB Collection	API Endpoint	Roles	Implementation Status
Chairperson Oversight	Cross-section project monitoring, user manager	`ChairpersonModule.tsx`, `ChairpersonFeasib.tsx`	`users`, `groups`, `proposals`	Client Firestore listeners	Chairperson	**Implemented**
Audit Trails	Immutable activity logging for compliance	`auditLogger.ts`, `AdviserAuditTrail.tsx`	`audit_logs`	`GET /api/audit/logs`	All	**Implemented**
Collaboration Chat	Team and adviser real-time messaging	`Messages.tsx`, `feasify-backend/index.js`	`messages`	Socket.IO (`io.on`)	All	**Implemented**

5. Architectural Constraints & Key Observations

1. **Dual Database Access Paths:** Feasify uses direct client-side Firestore SDK writes (`addDoc`, `updateDoc`) alongside Express 5 backend endpoints.
2. **Mapping Engine Reality:** Google Maps is not utilized; the platform uses **Leaflet v1.9.4 with OpenStreetMap Nominatim** and local landmark fallbacks (`philippineLandmarks.ts`).
3. **Hardcoded Role Bypass:** `chairperson@gmail.com` bypasses database role verification in both `ProtectedRoute.tsx` (line 31) and `feasify-backend/index.js` (line 97).
4. **Government APIs:** Public DTI BNRS and SEC eSPARC do not offer open unauthenticated APIs. Feasify accurately implements standardized sandbox mocking, trademark protection under RA 8293, and official portal deep-linking.